

Akmal Ashirmatov

akmalashirmatov@gmail.com | LinkedIn | OpenReview

M.Sc. Computer Science, ETH Zürich | LLM post-training, test-time scaling, agentic systems, multimodal generation

PUBLICATIONS & PREPRINTS

- **On Repulsive and Attractive Teachers: Separating Correctness from Behavior in Self-Distillation** [↗ FLLMPT @ NeurIPS 2026](#) 2026
 - Anton Baumann, **Akmal Ashirmatov**, Leo Schmidt-Traub, Frederike Lübeck, Jonas Hübötter, Thomas Kleine Buening, Andreas Krause
 - Studied attractive, repulsive, and contrastive self-distillation, showing that contrastive teachers cancel unintended behavioral shifts and improve reasoning.
- **Test-Time Scaling with Weak Verifiers via Self-Play** [↗ DL4C @ ICML 2026](#) 2026
 - Akira Yoshiyama*, **Akmal Ashirmatov***, Jonas Hübötter, Ido Hakimi, Andreas Krause
 - Self-play proposer-solver framework for improving code-generation models with weak verifier and environment feedback.
- **ESCAPE – Explaining Stable Diffusion via Cross Attention Maps and Prompt Editing** [↗ VISxAI 2025](#) 2025
 - Built an interactive system for cross-attention visualization and Prompt-to-Prompt-style localized image editing.

EXPERIENCE

- **SWE Intern, Applied Research team** Google, London Jun. 2026 – Sep. 2026
 - Prototyping video-generation capabilities for Gemini Live and contributing implementation code to the Gemini app codebase.
 - Authored a design document for integrating video generation as a new tool into the existing agent harness.
- **Research Intern, Zurich Research Center** Huawei Jun. 2025 – Sep. 2025
 - Worked on LLM inference optimization using Triton and vLLM; performed hardware profiling, ML compiler optimization, and IR-level debugging.

SELECTED PROJECTS

- **Distributed PEFT for Legal LLMs** [↗ Apertus 8B/70B](#) 2026
 - Built a PEFT/SFT pipeline with LoRA and full fine-tuning; ran multi-node Accelerate + DeepSpeed training with DDP/ZERO-2/ZERO-3.
- **Network Certification & Robustness** Reliable and Trustworthy AI 2026
 - Certified encoder/decoder properties via MILP (Gurobi) and DeepPoly relaxations; added PGD-based adversarial defense training.
- **3D Vehicle Detection from LiDAR Point Clouds** 1st place in class 2026
 - Redesigned second-stage RoI refinement with proposal-aligned canonical transforms, point-wise/geometric features, and IoU-aware scoring.
- **Music Semantic Search** 2025
 - Trained retrieval embeddings from CQT spectrograms using triplet loss with hard negative mining; an end-to-end cover-song retrieval pipeline.
- **Text Classification with Test-Time Augmentation & Selective Reasoning** [↗](#) 2025
 - Applied TTA via 4 paraphrases with weighted ensembling. Added selective LLM reasoning for uncertain cases

EDUCATION

- **ETH Zürich** M.Sc. Computer Science, GPA 5.93/6.0 Sep. 2024 – Apr. 2027 exp.
 - Selected coursework: Probabilistic AI, Advanced ML, Reliable and Trustworthy AI, Large-Scale AI Engineering, Computer Vision, Algorithms Lab.
- **Technical University of Munich** M.Sc. Data Engineering and Analytics, 1 semester Apr. 2024 – Sep. 2024
- **Lomonosov MSU Tashkent** B.Sc. Applied Mathematics & CS, with Honors, GPA 5.00/5.00 2018 – 2022

SKILLS

- **ML/LLM:** PyTorch, Hugging Face, vLLM, verl, GRPO/RLHF-style training, LoRA/PEFT, W&B
- **Systems:** Python, C/C++, Triton, Bash, Linux, Docker, Kubernetes, Ray, DeepSpeed, Spark, GCP, HPC/Slurm

AWARDS & COMPETITIONS

- ICPC Uzbekistan: 1st, 2nd, 1st, 1st place; 3-time National champion 2019 – 2022
- ICPC NERC: 2× 2nd-degree diploma; 4th and 5th in Central Asia 2021 – 2023
- Codeforces: Master, peak rating 2124 [↗](#)
- Meta Hacker Cup top 1.3%, Google Code Jam top 1.9%, CodeChef SnackDown top 0.8% 2021 – 2022
- Huawei iRunner: 6× prize-winner across 8 countries; high-performance heuristic optimization 2022 – 2025